Samyak Chakrabarty

PhD in Electrical and Computer Engineering

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Education

University of California, Los Angeles 3.91/4

Ms-PhD, Electrical and Computer Engineering, Advisor: Prof. Sudhakar Pamarti *2023 - Present*

Indian Institute of Technology, Kharagpur, India 9.39/10

B.Tech, Electronics and Electrical Communication Engineering *2019 - 23*

Projects

- **Attention Accelerator for VIT using DRAM-CIM Architecture** **Dec'24 - Present**

  Prof. Sudhakar Pamarti, University of California, Los Angeles (Master's Thesis)

  – Proposed a novel DRAM-based bit-cell architecture improving compute density, accuracy, and energy efficiency.
  – Designed and implemented circuit schematics and layouts, performing extensive Cadence Virtuoso analog simulations to verify power, area, and linearity across process corners.
  – Co-developed a system-level integration flow enabling full-network verification of accuracy, throughput, and power performance. Compared various trade-offs between system level and hardware choices.

- **CiM Hardware Accelerator for ResNet-18 in GF22FDSOI** **Sep'23 - Present**

  Prof. Sudhakar Pamarti, University of California, Los Angeles (DARPA Sponsored)

  – Developed system-level RTL architecture for interfacing with an Analog Compute-in-Memory (CIM) macro, and conducted comprehensive digital and mixed-signal verification to ensure functional correctness.
  – Designed a compact memory layout achieving 18Mb/mm^2 weight density, and proposed system-level optimization techniques to enhance inference accuracy and computational throughput.
  – Automated RTL-to-GDSII flow for mixed-signal SoC implementation in GlobalFoundries 22 nm technology, covering I/O integration, power planning, and floorplanning using TCL, Python, and OpenAccess scripting.
  – Integrated multiple IPs and optimized layout through detailed timing analysis and clock tree debugging, resolving complex data-path issues and achieving timing closure at 1 GHz for successful silicon tapeout.

- **A Light FPGA-based Accelerator for ANN Training** **Aug'22 - Apr'23**

  Prof. Anindya Sundar Dhar, Indian Institute of Technology, Kharagpur (Bachelor's Thesis)

  – Designed a Low-Energy Hardware Accelerator on FPGA in SystemVerilog and extensively evaluated timing and power on FPGA using Vivado.
  – Employed techniques like SIMD, massive parallelism, instruction pipeline to design a full-fledged processor. Handled IP modules for DSP and Block RAM.
  – Demonstrated 60% area reduction and 2x speedup compared to equivalent work, while maintaining high (95.18%) accuracy on MNIST.

- **Intuitive Design and Analysis Method for MPROs** **May'22 - Aug'22**

  Prof. Sudip Shekhar, University of British Columbia (Summer Internship)

  – Simulated Multipath Ring Oscillators (MPROs) on Virtuoso Cadence and defined an algorithm to classify them using parameters like ring and wiring modes, successfully identifying multiple prior structures as MPROs.
  – Performed extensive time-domain, frequency-domain, phase-noise and mismatch analyses on Virtuoso Cadence.
  – Validated the proposed classification and design methodology through post-layout simulations and silicon measurements on a TSMC 65nm test chip, confirming frequency stability and mode behavior.

- **Vt-Degradation Effect of Charge Trap Flash & Impact on ANN Training** **May'21 - Dec'21**

  Prof. Udayan Ganguly, Indian Institute of Technology, Bombay (Internship)

  – Developed mathematical model of Charge Trap Flash for RPU on MATLAB, based on experimental data, to predict behaviour when subjected to stochastic pulse train.
  – Analyzed the effects of Vt-degradation and other non-idealities of CTF on RPU training using the model, to determine optimum pulse width and height.
  – Designed SPICE model of CTF transistor and its programming/erasing mechanism using DC components like VCCS and capacitors incorporating the Vt-degradation effect.

Key Courses Taken

- Wireline & PLL*+
- RFIC & Wireless Circuits*
- Digital VLSI & EDA
- Data Converter Design*
- Digital Filter Design*+
- Signals & Systems'
- Network Theory'
- Algorithms'
- Linear Programming*
- DE & Vector Calculus
- Matrix Algebra
- Probability & Stochastics

*+Excellent Performance, *Graduate Course*

Technical Skills

Programming	SystemVerilog, Python, C/C++, TCL
Software	Cadence Design Suite (Innovus, Virtuoso, Genus, Calibre), Simvision, Matlab
Skills	RTL, Schematic, PnR, Flow Scripting, Git, Testbench Design, Timing Analysis

Publications

- M. Zeinali, V. Jacob, **S. Chakrabarty**, S. Pamarti. *An Energy and Area-Efficient 8b 300MS/s SAR ADC with Charge Trap Transistor-Based Calibrated DAC*. **JSSC 2026 (Submitted)** | Link
- G. Karfakis, **S. Chakrabarty**, V. Jacob, S, Qiao, S. Iyer, S. Pamarti, P. Gupta. *MXFormer: A Microscaling Floating Point Charge-Trap Transistor Compute-in-Memory Transformer Accelerator*. **GOMACTech 2026 | ASPLOS 2026 (Under Review)** | Link
- S. Qiao, S. Wang, J. Zhu, Z. Guo, M. Zeinali, V. Jacob, **S. Chakrabarty**, B. Vaisband, S. Iyer and S. Pamarti. *Back-gate Mediated VTH Mismatch Reduction to Enable 6-bit Precision in FD-SOI Charge-Trap Transistor*. **GOMACTech 2025** | Link
- S. Shrivastava, V. Saraswat, G. Dash, **S. Chakrabarty**, and U. Ganguly. *Non-ideal program-time conservation in charge trap flash for deep learning*. **SST 2023** | Link

Awards and Achievements

- **Best Poster, IBM AI Symposium 2025**, Among 150+ Student posters *Nov'25*
- **Department Fellowship**, Recipient among 300+ students at ECE, UCLA. *Apr'23*
- **Best B.Tech Project Work**, Recipient among 200+ students in ECE, IIT Kharagpur. *Sep'23*
- **DAAD WISE**, Funded Research Fellowship in Germany among top 130 students *Jan'22*
- **MITACS**, Funded Research Fellowship in University of British Columbia, Vancouver *Mar'22*

Leadership

- **Former Design and Media Subhead**, Spring Fest IIT Kharagpur
- **Senior Editor (Design)**, Technology Literary Society, IIT Kharagpur
- **Governor**, Spectra Fine Arts Society, Kharagpur